With Provisioned Throughput mode enabled, you should keep an eye on the utilization of the throughput that you provisioned. This allows you to increase the provisioned throughput to avoid performance problems caused by EFS becoming the bottleneck of your system. Listing 9.9 shows how to create a CloudWatch alarm to monitor the utilization of the maximum throughput of an EFS filesystem. The same CloudWatch alarm works for filesystems with Bursting Throughput mode as well. This combines the metrics `MeteredIOBytes`, the throughput to the filesystem, and `PermittedThroughput`, the maximum throughput of the filesystem, by using metric math (see http://mng.bz/qdvA for details).

Listing 9.9 Monitoring using a CloudWatch alarm and metric math

```
PermittedThroughputAlarm:
  Type: 'AWS::CloudWatch::Alarm'
  Properties:
    AlarmDescription: 'Reached 80% of the permitted throughput ...'
    Metrics:                                          ①
    - Id: m1                                          ②
      Label: MeteredIOBytes
      MetricStat:
        Metric:
          Namespace: 'AWS/EFS'
          MetricName: MeteredIOBytes                  ③
          Dimensions:
          - Name: FileSystemId
            Value: !Ref FileSystem
        Period: 60
        Stat: Sum
        Unit: Bytes
      ReturnData: false
    - Id: m2
      Label: PermittedThroughput
      MetricStat:
        Metric:
          Namespace: 'AWS/EFS'
          MetricName: PermittedThroughput             ④
          Dimensions:
          - Name: FileSystemId
```